

PF-FDTD Sheath study Progress and Analysis Report

Antenna Impedance in Plasma and Tu (2008) Sheath Validation

Abstract

This report records the experimental campaign on the Plasma Fluid Finite-Difference Time-Domain (PF-FDTD) solver from the February 2026 dipole-in-plasma baseline through the August 2026 dense continuous-wave (CW) sheath-width sweep. The scientific goal is to reproduce the Tu (2008) result that a vacuum sheath of increasing width S_d around a radio-frequency dipole raises the series resonance f_{res} toward the free-space value. Early pulse and low-frequency CW campaigns either failed as measurements or showed Z independent of S_d . A coupling bug—sheath depletion seeded from the plasma-on mask before antenna geometry existed—placed the density hole at the domain edge rather than next to the wire. After a PEC-seeded fix, feed impedance depends strongly on sheath width. A dense 66-case CW sweep at $f_p = 2$ MHz (1.50–2.30 MHz) completed on 18 August 2026: $f_{\text{res}}(S_d = 0) \approx 1.846$ MHz, but no $\text{Im}\{Z\}$ + to $-$ crossing appears for $S_d \geq 2$ in that band, so the Tu upward shift was not reproduced with this observable.

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Chapter 1

Project Goal

1.1 The solver

Plasma Fluid Finite-Difference Time-Domain (PF-FDTD) is a three-dimensional electromagnetic solver coupled to multi-species plasma fluid equations. It originated in Jeffrey D. Ward’s plasma-frequency-probe work at Utah State University [2] and has been modernized with a modular `src/` layout, OpenMP, Python visualization, and scripted experiment drivers.

On a staggered Yee grid the code advances Maxwell’s equations

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad (1.1)$$

$$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \partial_t \mathbf{E} + \mu_0 \mathbf{J}, \quad (1.2)$$

with plasma current $\mathbf{J} = \sum_s q_s n_s \mathbf{u}_s$ from continuity and momentum updates for each species. Antenna geometry and sources are read from `.str` input files. The feed observables used throughout this report are the voltage $V(t)$ and current $I(t)$ written to `.vc` files; input impedance is

$$Z(f) = \frac{V(f)}{I(f)}. \quad (1.3)$$

1.2 Scientific target: Tu (2008)

The active research goal is to reproduce the sheath-induced resonance shift reported by Tu [1]. A conducting antenna in plasma develops a vacuum (or strongly depleted) sheath of width S_d cells next to the conductor. Inside that layer the ambient density is essentially zero; outside it recovers to the bulk N_0 .

Increasing S_d reduces dielectric loading near the antenna, so the series resonance f_{res} —the frequency where $\text{Im}\{Z\} = 0$ with a $+$ to $-$ crossing—should move *upward* toward the free-space value.

An equivalent-circuit picture of the same effect is a sheath capacitance in series with the antenna feed:

$$Z_{\text{in}}(f) = Z_{\text{ant}}(f) + \frac{1}{j\omega C_{\text{sh}}}, \quad C_{\text{sh}} \sim \frac{\varepsilon_0 A}{S_d \Delta x}. \quad (1.4)$$

In FDTD this is realized as a volumetric vacuum gap on Yee nodes adjacent to the conductor: depleted N_0 implies $J \approx 0$, so those cells act as vacuum capacitors loading the feed. Tu’s lumped C_{sh} is the circuit limit of that gap; it is not implemented as a separate circuit element.

1.3 Campaign protocol

The planned observable is $f_{\text{res}}(S_d)$ for $S_d \in \{0, 2, 4, 6, 8, 10\}$ cells on a $70 \times 70 \times 65$ dipole grid with $\Delta x = 0.04$ m. The same geometry appears in every chapter of this report: a z -directed thin-wire dipole from $(35, 35, 20)$ to $(35, 35, 45)$ with a hard E_z source at $(35, 35, 33)$.

What changed between phases is the excitation (broadband pulse versus CW sine), the plasma frequency, and—after August 2026—whether the sheath was actually attached to the conductor. Each subsequent chapter records what was tried, what succeeded, and what failed.

1.4 How to read this document

Chapters are chronological. Chapter 2 is the February 2026 free-space versus plasma baseline on this dipole; it is a prerequisite, not a detour. Chapters 3–5 are pre-fix sheath campaigns. Chapter 6 diagnoses the coupling bug. Chapters 7–8 are post-fix confirmation and the dense CW Tu sweep completed 18 August 2026.

Chapter 2

February Baseline: Dipole in Free Space versus Plasma

2.1 What we tried

After the January 2026 modernization (modular sources, CMake, OpenMP, Python plotting), the first physics question was whether the rebuilt solver could show plasma loading of a dipole at all. That is the prerequisite for later asking whether a vacuum sheath *changes* that loading.

All February runs used the same antenna that later became **sheath.str**: $70 \times 70 \times 65$ cells, $\Delta x = 0.04$ m, PEC from (35, 35, 20) to (35, 35, 45), feed at (35, 35, 33), type-5 differentiated Gaussian. The only material difference from the May sheath input is the source parameter (100 MHz here versus 30 kHz later).

Table 2.1: February 2026 dipole runs under **results/**.

Run	Question	Settings	Outco
production_run_8000	Does OpenMP plasma run?	$f_p = 5.3$ MHz, $f_c = 1.43$ MHz, 8k steps	Yes: V
freespace_test	Free-space dipole	no plasma CLI args, 100k steps	Baseli
ionoplasma_test	Plasma on, same dipole	$f_p = 0.5$ MHz, $f_g = 1.1$ MHz, 100k steps	Plasm
ionoplasma_test_5_1_11	Collision / cyclotron variant	$f_p = 0.5$ MHz, $f_c = 0.05$ MHz, 64k steps	Param
benchmarks	Legacy vs new OpenMP	dipoleTest, 10k steps	Timin
test_freespace / test_plasma	CLI plasma-flag check	verify_plasma_flag.bat	Incom

2.2 What succeeded

The modernized OpenMP executable ran to completion. Free-space and plasma cases produced distinct voltage/current traces and impedance spectra (Figures 2.1–2.2). The Python $Z(f) = V(f)/I(f)$ pipeline used in every later chapter starts here. OpenMP scaling was recorded in Figure 2.3.

This established that *plasma loading of feed impedance is measurable* on this grid. Without that result, a sheath-width sweep would have had no baseline.

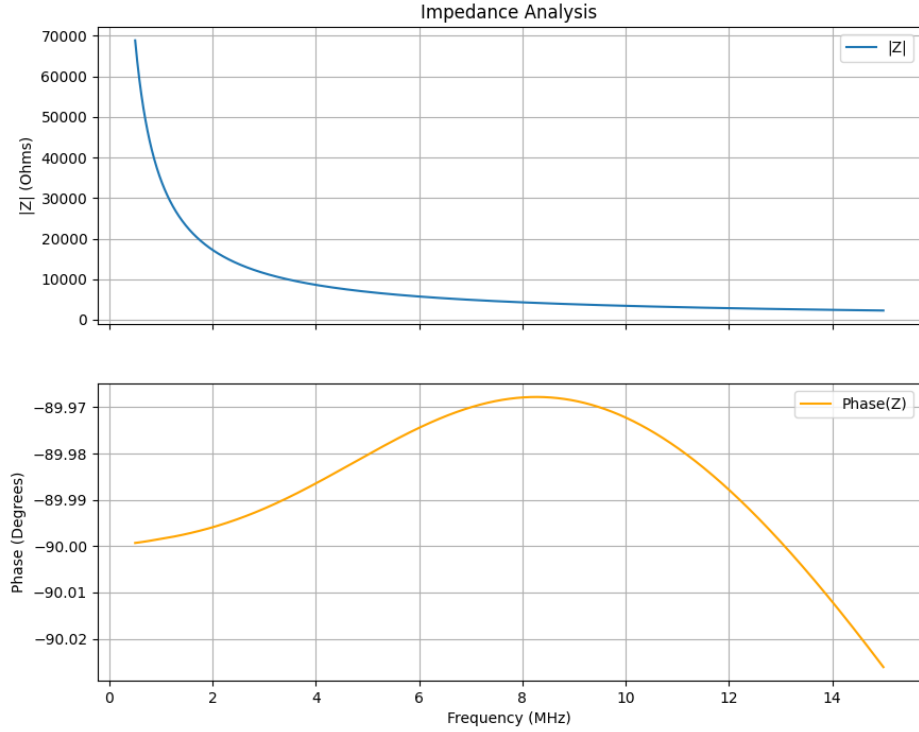


Figure 2.1: Free-space dipole impedance from `results/freespace_test` (February 2026). Same geometry as all later sheath runs.

2.3 What failed or was left open

- There was no sheath parameter S_d . Density was spatially uniform.
- Plasma frequencies (5.3 MHz or 0.5 MHz) were Ward-like ionospheric defaults, not yet the Tu-oriented 200 kHz then 2 MHz choices.
- The 100 MHz differentiated Gaussian is a broadband pulse. Frequency resolution and spectral artifacts from this class of drive reappear as a measurement failure in Chapter 3.
- Several February runs included a cyclotron frequency (background $\mathbf{B}$). Later sheath work set $f_{\text{cyc}} = 0$.
- The CLI plasma-flag stubs (`test_freespace`, `test_plasma`) did not produce usable logs.

The May sheath campaign is the next chapter of this same antenna-in-plasma story: keep the dipole, add a vacuum layer of width S_d , and look for an upward shift of f_{res} .

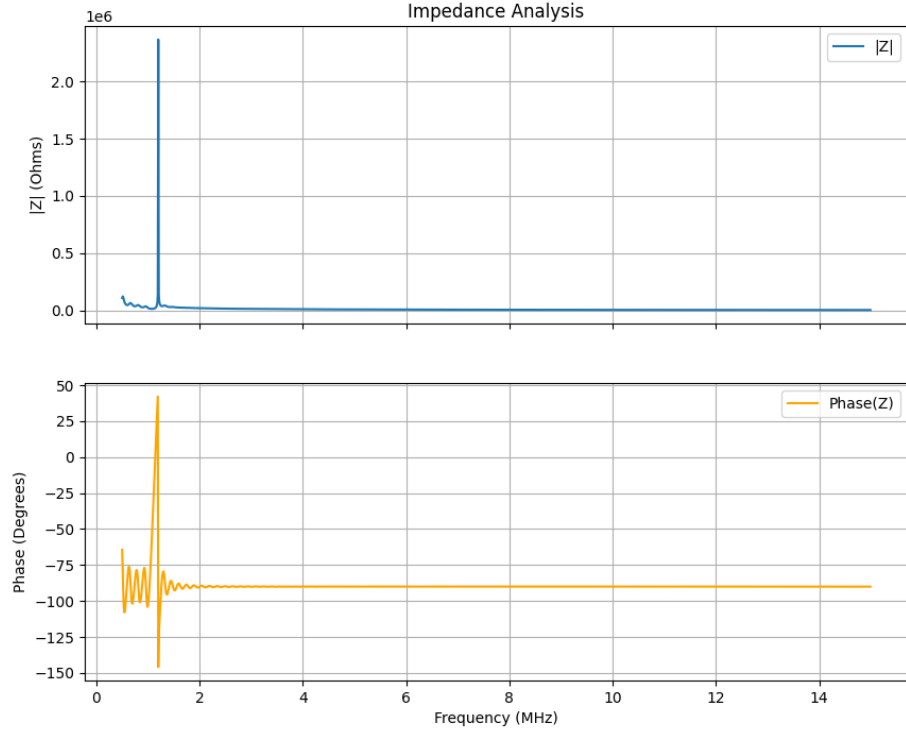


Figure 2.2: Plasma-loaded dipole impedance from `results/ionoplasma_test` ($f_p = 0.5$ MHz). Plasma changes $Z(f)$ relative to Figure 2.1.

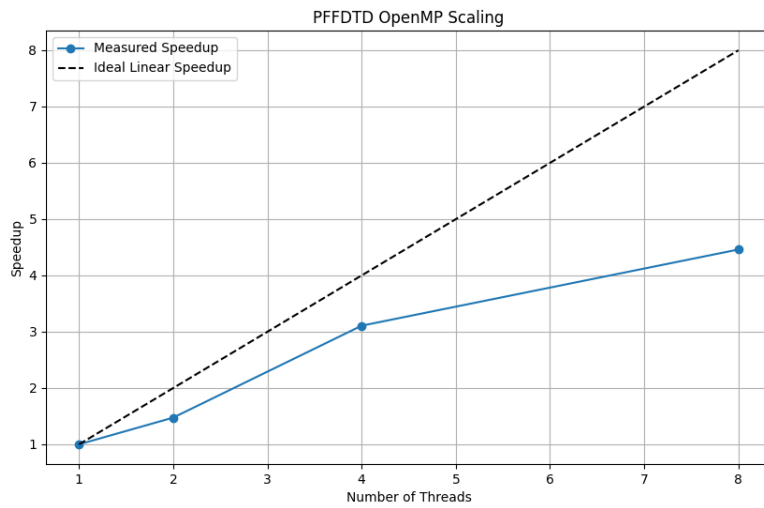


Figure 2.3: OpenMP scaling recorded during the February solver checkout.

Chapter 3

Pulse Sheath-Width Sweep

3.1 What we tried

The first sheath campaign (13–16 May 2026) added a step-profile vacuum sheath of width S_d cells around the February dipole and drove it with a broadband pulse. The protocol is in `docs/sheath/SHEATH_VALIDATION`.

- Input: `sheath.str` — source type 5 (differentiated Gaussian) at 30 kHz.
- Plasma: $f_p = 200$ kHz, $T = 2000$ K, no background **B**.
- Sheath: $S_d = 0, 2, 4, 6, 8, 10$.
- Analysis: Hanning-window FFT of the full $V(t)/I(t)$ traces (`scripts/analyze_sheath_results.py`).
- Data: `results/sheath_sweep/`, plus a longer $S_d = 0$ attempt in `results/sheath_long/`.

The expected signature was a monotonic increase of f_{res} with S_d .

3.2 What failed

No $\text{Im}\{Z\} = 0$ crossing was found below 1 MHz. Fallback $|Z|$ peaks fell in the multi-gigahertz band (Table 3.1) and are sampling/FFT artifacts, not the kHz/MHz antenna resonance discussed by Tu. A long-trace reanalysis of `sheath_long/sd0` with a 200 kHz cap still found no zero-crossing (fallback peak 46.8 kHz).

Table 3.1: Pulse-FFT “resonance” picks from the May S_d sweep. All $\text{Im}\{Z\}$ crossings in the analysis band are missing; GHz peaks are not physical.

S_d (cells)	$\text{Im}\{Z\}$ zero-crossing	Fallback $ Z $ peak (Hz)
0	—	6.61×10^9
2	—	6.16×10^9
4	—	5.28×10^9
6	—	5.28×10^9
8	—	5.28×10^9
10	—	5.28×10^9

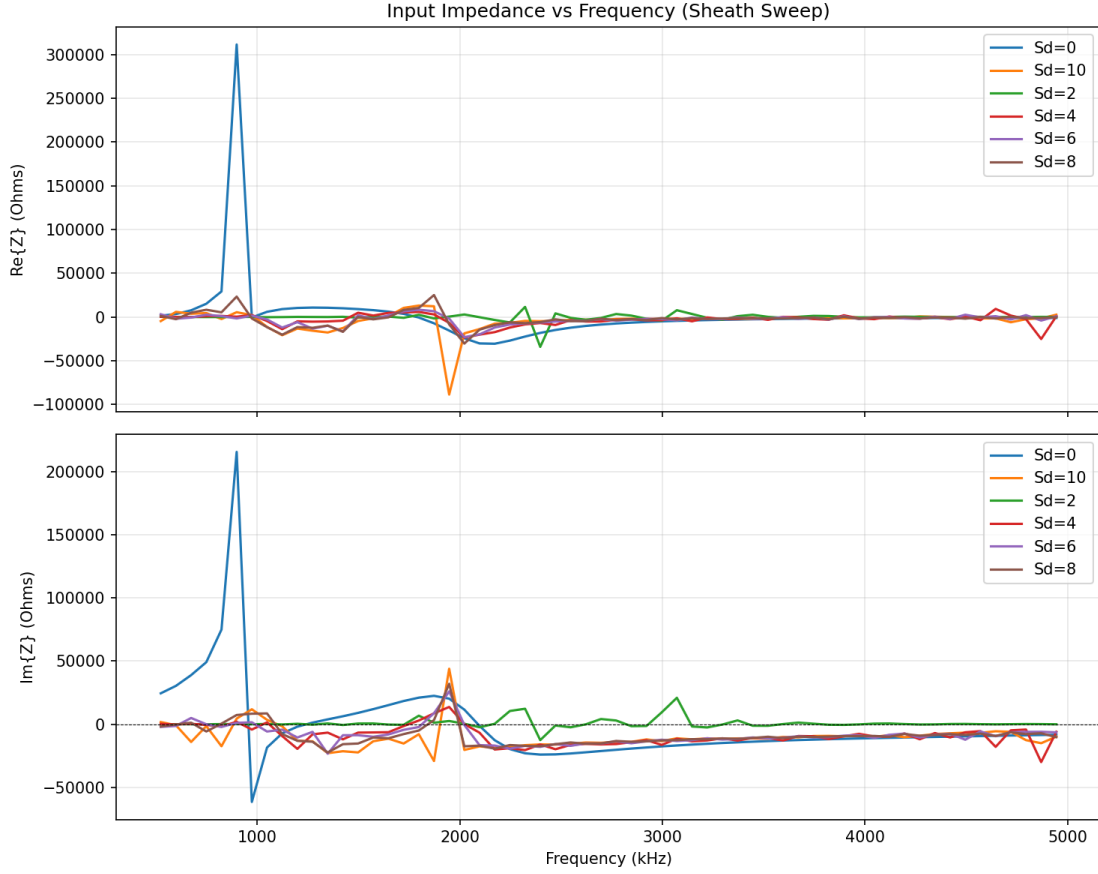


Figure 3.1: Broadband pulse FFT impedance versus frequency for $S_d = 0 \dots 10$. No usable series resonance in the Tu band.

3.3 What succeeded, narrowly

All six S_d runs completed and wrote readable `data.vc` files. The analyzer and sweep scripts became the template for later campaigns. That is an infrastructure success, not a physics result.

3.4 Interpretation

This campaign *cannot test* Tu (2008): the observable f_{res} was never resolved. Possible contributors, recorded at the time, were coarse DFT bins at kHz scales, transient-dominated traces, and a drive that did not sit on the resonance. A later finding (Chapter 6) adds a second, independent failure: even a perfect FFT would have seen no S_d dependence, because the sheath was not coupled to the feed.

Policy after this phase: do not spend further wall time on long broadband pulse runs until the measurement method is confirmed with narrow-band or CW excitation.

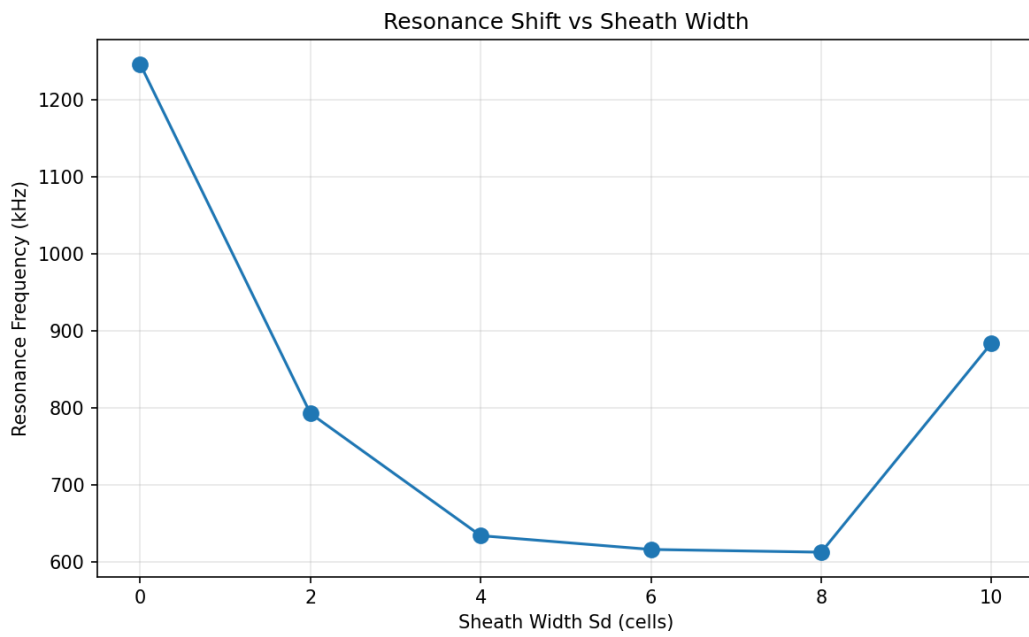


Figure 3.2: Apparent resonance versus S_d from the pulse analyzer. The GHz-scale picks are FFT artifacts; this campaign cannot test Tu (2008).

Chapter 4

Narrow-Band and Two-Dimensional CW below f_p

4.1 What we tried

Because the pulse FFT did not resolve f_{res} , the next phase switched to a continuous-wave sine (`sheath_sine.str`) and extracted a phasor $Z = V/I$ from the last half of each trace. Two related campaigns ran from late May through 11 August 2026.

Narrow-band frequency scan. `results/sheath_narrow_band/`, driver `scripts/run_sheath_narrow_band.ps`. Sixteen frequencies from 10 kHz to 400 kHz. Folder names are `sd{Hz}` (for example `sd20000`): a PowerShell expansion bug treated S_d as empty, so these directories are frequencies, not sheath widths. The default S_d was 0.

Two-dimensional $S_d \times f$ sweep. `results/sheath_2d_sweep/`. $S_d = 0$ at 20–400 kHz; $S_d = 2, 4, 6, 8, 10$ at 100, 150, 200, and 300 kHz. Phasor analysis: `scripts/analyze_2d_sweep.py`.

Both campaigns sit well below the ~ 1.8 MHz resonance later found near $f_p = 2$ MHz.

4.2 What succeeded

CW phasor impedance is a stable measurement. Traces reach a usable steady state, and $\text{Re}\{Z\}$ and $\text{Im}\{Z\}$ versus drive frequency are smooth enough to plot (Figures 4.1–4.2). This is the measurement method used for all subsequent screens.

4.3 What failed

The physics test failed, for two stacked reasons.

1. **Wrong band.** These frequencies are far below the series feature later located near 1.78 MHz (Chapter 5). A null result here cannot confirm or refute Tu.
2. **No S_d dependence even where compared.** Tables for $S_d = 2, 4, 6, 8, 10$ at 100–300 kHz are numerically identical to many digits. Example at 200 kHz: $S_d = 0$ gives $\text{Re}\{Z\} \approx 7841 \Omega$, $\text{Im}\{Z\} \approx 3046 \Omega$; $S_d = 10$ agrees to the reported precision. The sheath never loaded the feed.

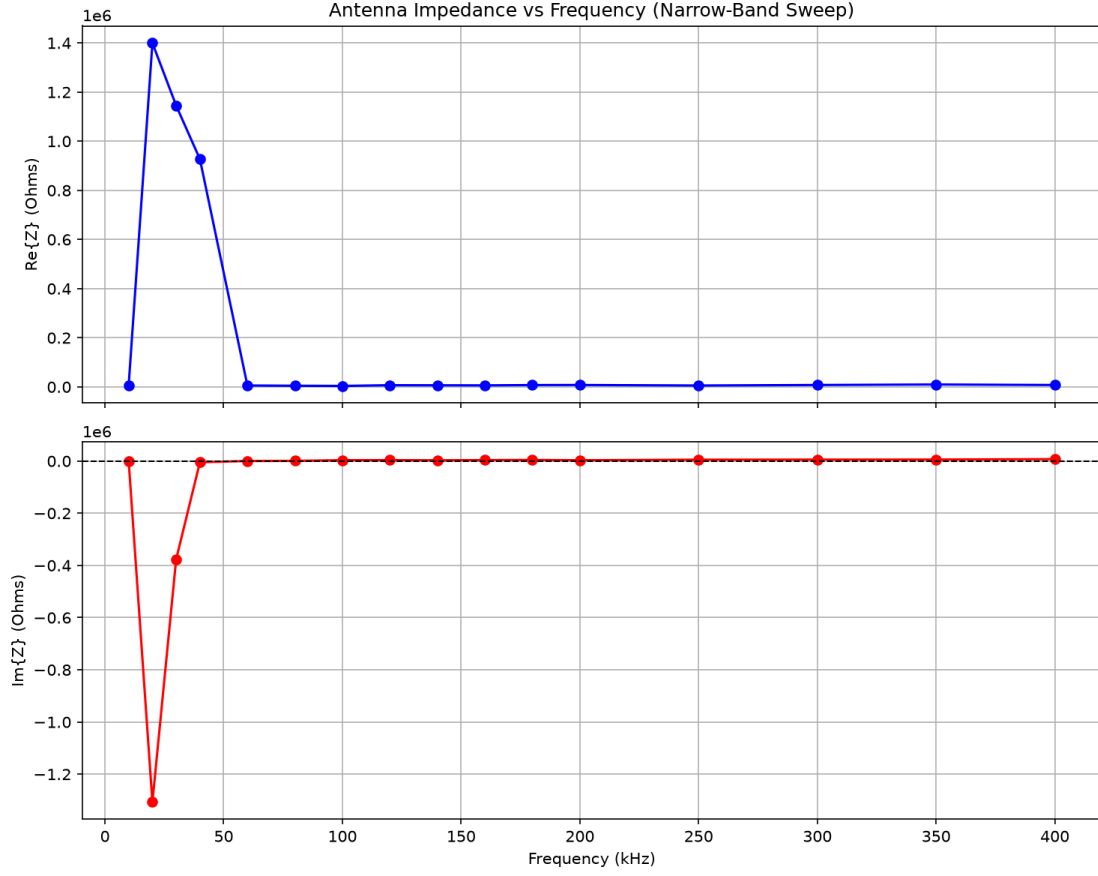


Figure 4.1: Narrow-band CW impedance, 10–400 kHz. Folders are misnamed `sd{Hz}`; the physical sheath width is the script default $S_d = 0$.

The overlapping S_d curves in Figure 4.2 are the first clear sign that the volumetric sheath was not attached to the antenna. That conclusion is made definitive by the f_p -screen in the next chapter, which repeats the comparison *on* the resonance.

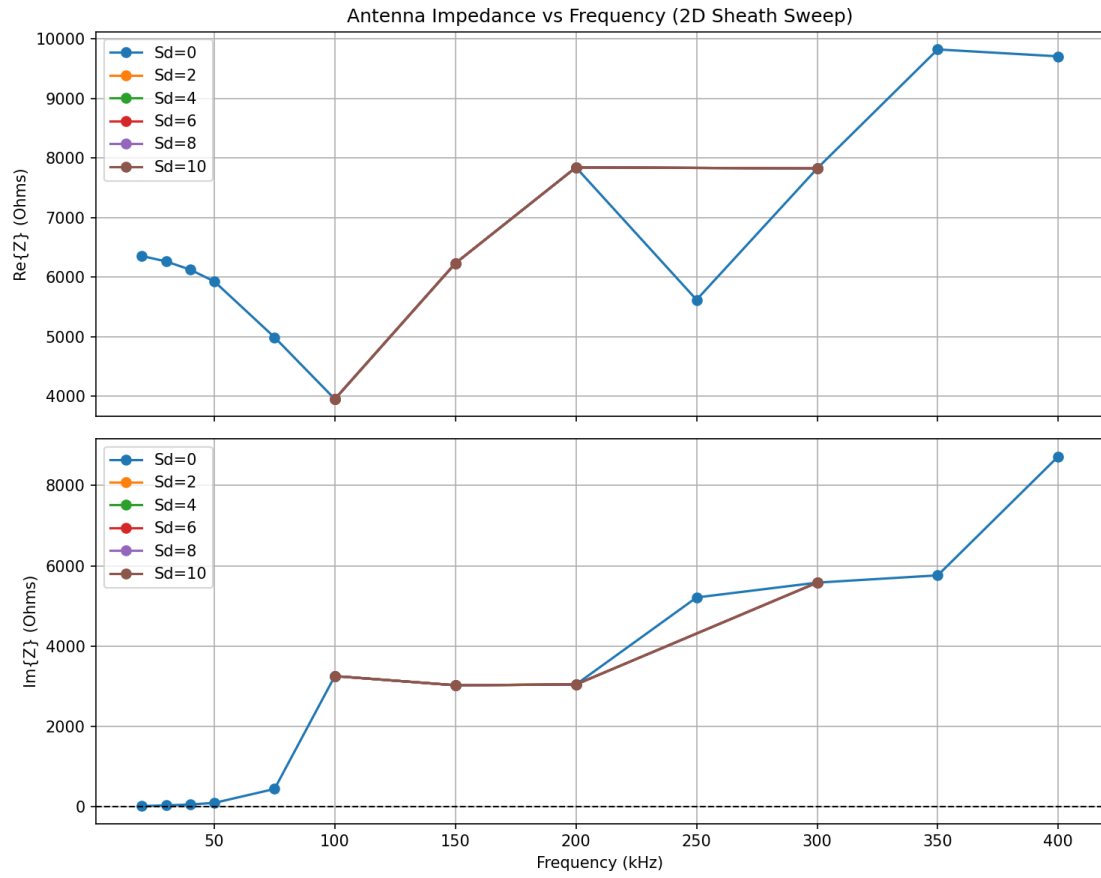


Figure 4.2: Two-dimensional CW sweep at 20–400 kHz. Curves for $S_d = 2 \dots 10$ overlaid one another; sheath width does not enter the feed.

Chapter 5

CW Plasma-Frequency Screen

5.1 What we tried

On 11–12 August 2026 a CW screen was run at $f_p = 2$ MHz for $S_d = 0$ and $S_d = 10$ only, with drive frequencies spanning 0.6–3.0 MHz so that the series resonance could not be missed. Driver: `scripts/run_sheath_fp_screen_parallel.ps1`. Data: `results/sheath_fp_screen/` (20 cases, 8.65 h, all completed). Analysis again used steady-state phasor Z on the last 50% of each trace.

5.2 What succeeded as a measurement

A clean series resonance appears at $f_{\text{res}} \approx 1.775$ MHz for *both* sheath widths: $\text{Im}\{Z\}$ is positive at 1.7 MHz and negative at 1.9 MHz. The CW method works. This is the first time the campaign resolved the observable Tu requires.

5.3 What failed as physics

Impedance agrees to $\sim 0.003\%$ between $S_d = 0$ and $S_d = 10$ at every frequency (Table 5.1, Figure 5.1). Tu requires an *upward* shift of f_{res} with sheath width. Identical $Z(f)$ means the sheath never enters the feed. This folder is the pre-fix negative control and should be kept.

Table 5.1: Pre-fix CW f_p -screen. $S_d = 0$ and $S_d = 10$ are indistinguishable.

f (MHz)	$\text{Re}\{Z\} S_d=0$	$\text{Re}\{Z\} S_d=10$	$\text{Im}\{Z\} S_d=0$	$\text{Im}\{Z\} S_d=10$
0.60	13414.2	13413.9	+9035.6	+9035.4
1.00	19010.1	19009.5	+4516.7	+4517.0
1.40	23192.8	23193.0	+7124.8	+7125.1
1.70	39980.4	39980.9	+9845.3	+9845.2
1.90	70942.1	70942.3	−16284.5	−16284.7
2.00	56193.4	56193.0	−55242.0	−55242.7
2.10	21336.2	21336.4	−53151.9	−53151.6
2.20	9163.7	9163.6	−40864.8	−40864.8
2.40	3312.4	3312.5	−29552.6	−29552.6
3.00	566.8	566.9	−16669.6	−16669.6

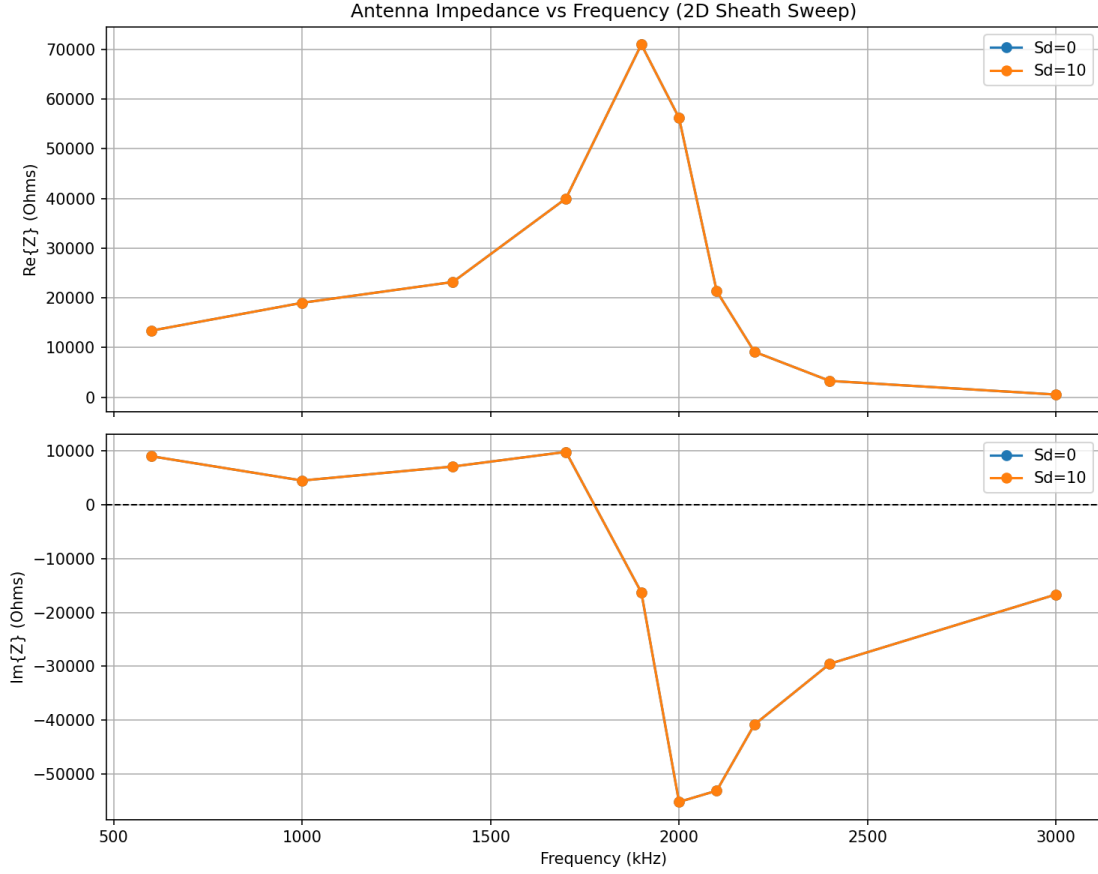


Figure 5.1: CW f_p -screen impedance. $S_d = 0$ and $S_d = 10$ overlap at every point; $f_{\text{res}} \approx 1.775$ MHz for both.

5.4 Interpretation

The measurement problem of Chapter 3 is solved. The remaining problem is in the model: a density hole exists somewhere, but not where V and I are formed. Chapter 6 dumps N_0 along a line through the feed and finds that hole at the domain edge.

Chapter 6

Diagnosis and Coupling Fix

6.1 How the sheath was supposed to enter the model

The active path is `src/physics/plasma.cpp`. Intended sequence:

1. Initialize `N0_SPATIAL` to uniform bulk N_0 .
2. Set `SIG = 1` where plasma couples into Maxwell (`Ecalcmod: -Cμ SIG J`).
3. If $S_d > 0$, dilate a distance field from a seed and set

$$N_{0,\text{spatial}} = N_0 \cdot N_{\text{MIN_RATIO}} \quad (N_{\text{MIN_RATIO}} = 10^{-6})$$

for $0 < d \leq S_d$.

4. Fluid updates and current J use `N0_SPATIAL`. Depleted cells give $J \approx 0$.

Feed observables (`src/source/source.cpp`) are a hard E_z voltage at (35, 35, 33) and an Ampère loop of B_x, B_y around that cell. Plasma loading of Z therefore requires depleted response in *near-antenna* Yee nodes—not at the outer domain wall.

6.2 Three overlapping mistakes

Init order. `pffddtd.cpp` called `PLASMAclear()` *before* `setup2()`. At that moment `ClearArrays()` had set `ERX = ERY = ERZ = 1` everywhere. The “mark antenna / turn plasma on” loop therefore set `SIG = 1` on the entire interior plasma box, not on the dipole.

Wrong sheath seed. Sheath dilation seeded from `SIG > 0.5`. Even after reordering geometry, `SIG = 1` is defined where *any* of `ERX/ERY/ERZ == 1`. The thin-wire dipole in `sheath.str` is only `ERZ = 0` at (35, 35, 20–45); `ERX` and `ERY` stay 1. So `SIG` is a plasma-on mask, not a conductor mask. Distance $d = 0$ filled the whole interior; depletion landed in an S_d -cell shell at the *domain boundary* (~ 0.4 m for $S_d = 10$, $\Delta x = 0.04$ m), while the dipole sits at the grid center.

Overloaded SIG. `SIG` was used as (a) plasma–Maxwell coupling, (b) sheath seed, and (c) antenna density sink in `NBCcalc`. Those three meanings conflict. Near-antenna nodes kept bulk N_0 , so V and I were independent of S_d .

6.3 Diagnostic: N_0 along a line normal to the dipole

A two-iteration dump of N0.SPATIAL along i at $j = 35$, $k = 33$ (feed plane, $+x$ normal to the z -dipole) is in `results/n0_diag/`.

Before the fix (legacy SIG-seeded binary). Both $S_d = 0$ and $S_d = 10$ show bulk $N_0 = 4.96 \times 10^{10} \text{ m}^{-3}$ next to the wire ($i \approx 35$). For $S_d = 10$, depletion to $N_0 \times 10^{-6}$ appears only near the domain edge ($i = 1 \dots 5$), outside the plasma box. A density hole exists, but the feed does not use those cells.

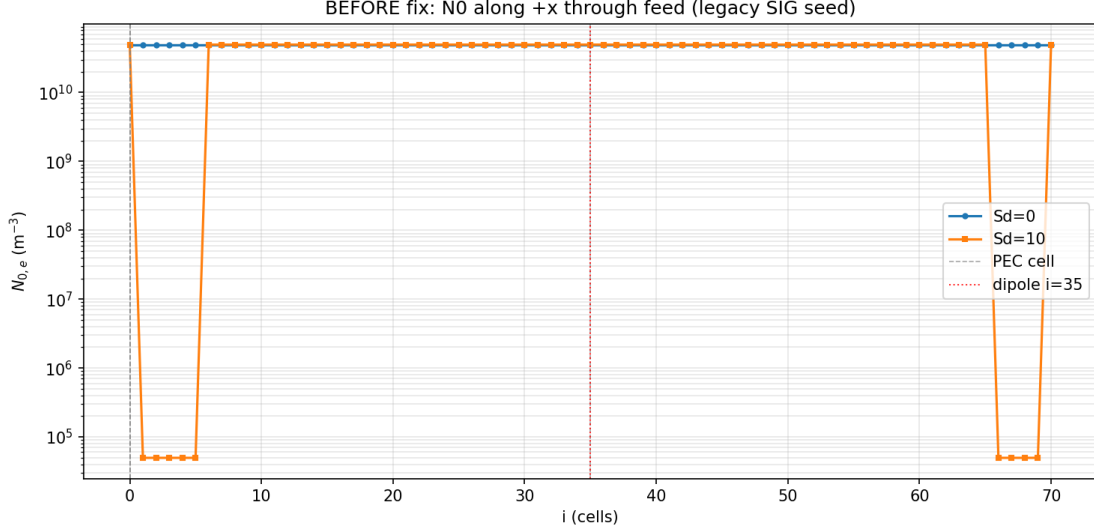


Figure 6.1: N_0 along a line through the feed, before the coupling fix. $S_d = 10$ depletes the domain edge, not the wire.

After the fix. Sheath seeds from 26 dipole PEC cells (not $\sim 14\text{k}$ false boundary PECs). $S_d = 0$ is uniform bulk. $S_d = 10$ depletes for $0 < |i - 35| \leq 10$; the domain edge remains bulk. Near-wire median N_0 : 4.96×10^{10} ($S_d = 0$) versus 4.96×10^4 ($S_d = 10$).

6.4 What changed in the code

1. Reorder init in `pffddtd.cpp`: `ClearArrays` $\rightarrow$ `setup2` (antenna ER*) $\rightarrow$ `PLASMAclear` $\rightarrow$ `ApplySheath` $\rightarrow$ `DumpN0Line` $\rightarrow$ `Ninitall`.
2. Seed the sheath from PEC cells (`ERX == 0` or `ERY == 0` or `ERZ == 0`), not from SIG.
3. Keep SIG = 1 as plasma–Maxwell coupling only, including sheath cells. With depleted N_0 , $J \approx 0$, so those cells act as vacuum.
4. Restrict the `NBCcalc` density sink to PEC cells.
5. Initialize the full $0 \dots s_x$ index range in `ClearArrays` so uncleared $i/j/k = 0$ faces do not look like PEC ($\sim 14\text{k}$ bogus seeds $\rightarrow$ 26 real dipole cells).

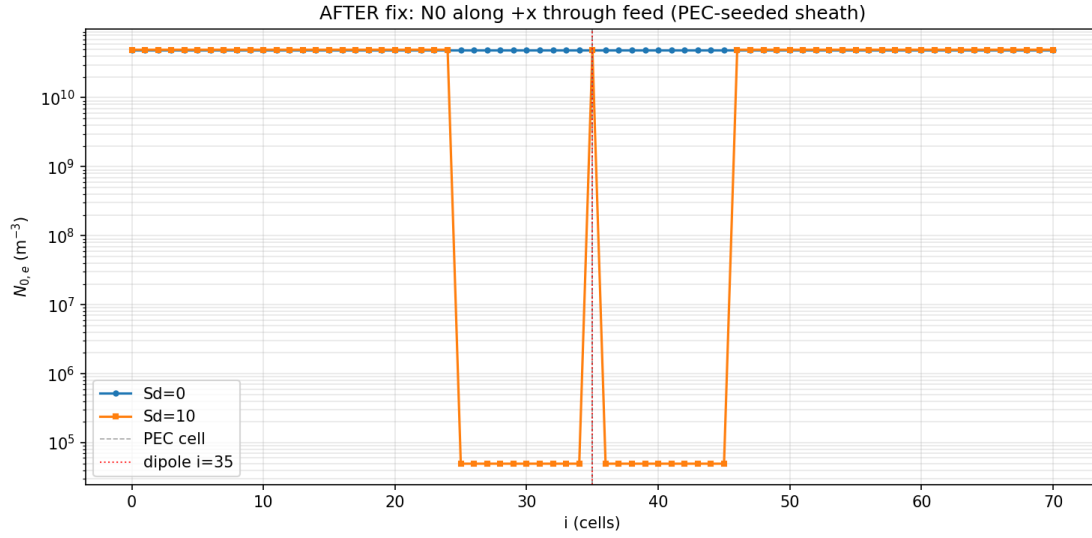


Figure 6.2: N_0 after the PEC-seeded fix. Depletion sits next to the wire for $S_d = 10$.

6. CLI `MaxIter` is re-applied after `setup1` reads the `.str` fail-safe line (previously ignored).
7. Path buffers enlarged so long `results/...` paths do not overflow.

Legacy SIG-seeded dumps can be reproduced with `-DSHEATH_LEGACY_SIG_SEED=1`. The next chapter confirms that feed impedance now depends on S_d .

Chapter 7

Post-Fix Confirmation

7.1 CW confirm at the resonance shoulder

A short CW check at 1.7 MHz and 1.9 MHz, $S_d = 0$ versus $S_d = 10$, $f_p = 2$ MHz, $\text{MaxIter} = 100000$, is in `results/sheath_confirm.z/`.

Table 7.1: Post-fix CW confirm. Relative change $|\Delta Z|/|Z_0|$ was $\sim 0.003\%$ before the fix.

f (MHz)	$\text{Re}\{Z\}$ $S_d=0$	$\text{Re}\{Z\}$ $S_d=10$	$\text{Im}\{Z\}$ $S_d=0$	$\text{Im}\{Z\}$ $S_d=10$	$ \Delta Z / Z_0 $
1.70	39564	6406	+10023	-21936	112.8%
1.90	71059	4573	-15542	-22129	91.9%

Succeeded. Impedance now depends strongly on sheath width. $S_d = 0$ still shows an $\text{Im}\{Z\}$ sign change between 1.7 and 1.9 MHz ($f_{\text{res}} \sim 1.78$ MHz). $S_d = 10$ is strongly capacitive at both frequencies: the feed sees the vacuum sheath. The coupling bug is resolved.

Still open. Two frequencies cannot locate the new f_{res} for every S_d . A dense CW grid is required for a Tu shift curve (Chapter 8).

7.2 Post-fix long pulse FFT

A full S_d grid after the fix is in `results/sheath_pulse_long/` ($f_p = 2$ MHz, $\text{Spar} = 2$ MHz, $\text{MaxIter} = 2 \times 10^5$, $\text{VcRate} = 50$). Wall time 6.57 h; all six cases exited 0. Analyzer: `scripts/analyze_pulse_long.py`, using CW-compatible + to - crossings in 1.2–3 MHz.

Succeeded. Pulse FFT now *sees* S_d . $S_d = 10$ capacitive at 1.7–1.9 MHz matches the CW confirm.

Failed as a precision tool. Stock - to + resonance picks are often spurious. Even with + to - crossings, $df \sim 75$ kHz is too coarse to map a clean $f_{\text{res}}(S_d)$ curve. A denser CW sweep is the remaining measurement.

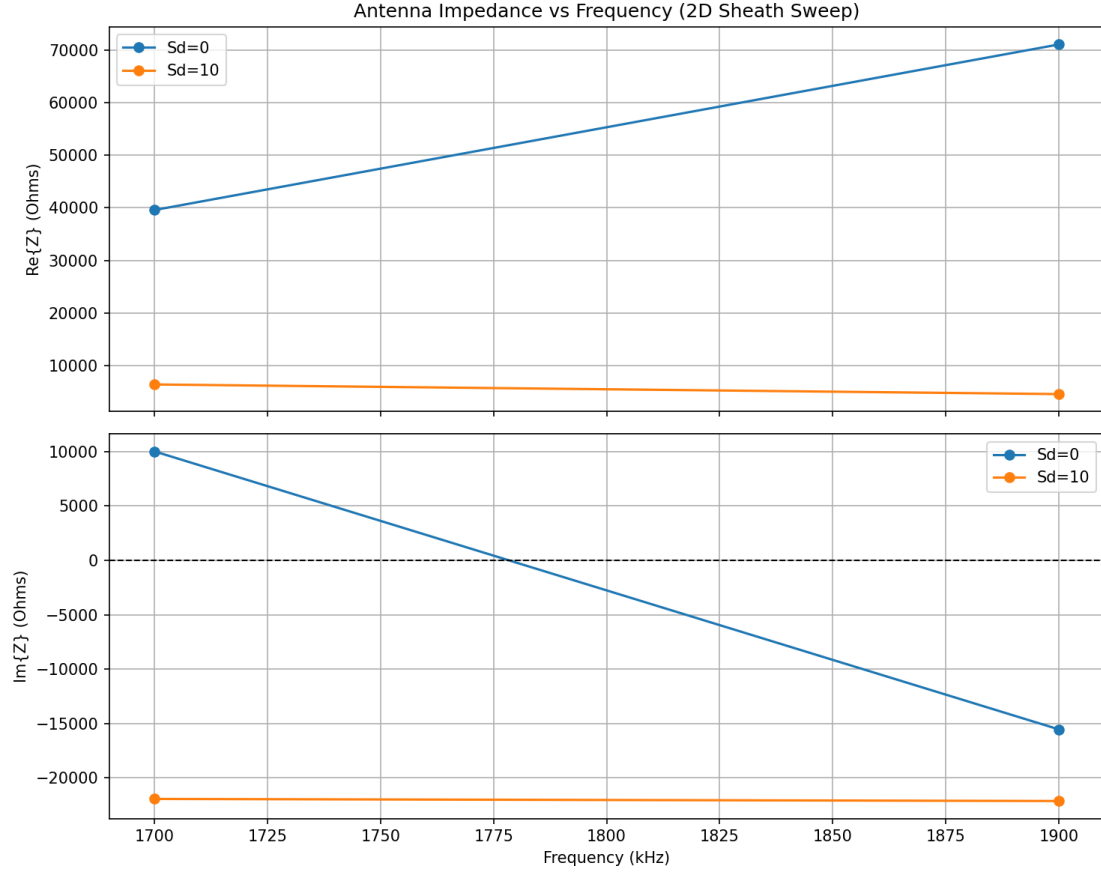


Figure 7.1: Post-fix CW impedance at 1.7 and 1.9 MHz. $S_d = 0$ still crosses $\text{Im}\{Z\} = 0$ between the two points; $S_d = 10$ is capacitive at both.

Table 7.2: Post-fix pulse FFT. Automatic picks are noisy; $df \sim 75$ kHz.

S_d	f_{res} (MHz), $\text{Im} \rightarrow -$ in 1.2–3 MHz	$\text{Im}\{Z\}$ at 1.7 MHz	$\text{Im}\{Z\}$ at 1.9 MHz
0	2.09	+18325	+22444
2	1.23 (noisy; also 2.03)	−536	+1115
4	2.02	−1542	+8541
6	2.02	−4527	+8638
8	2.00	−7910	+3205
10	2.00	−15393	−29243

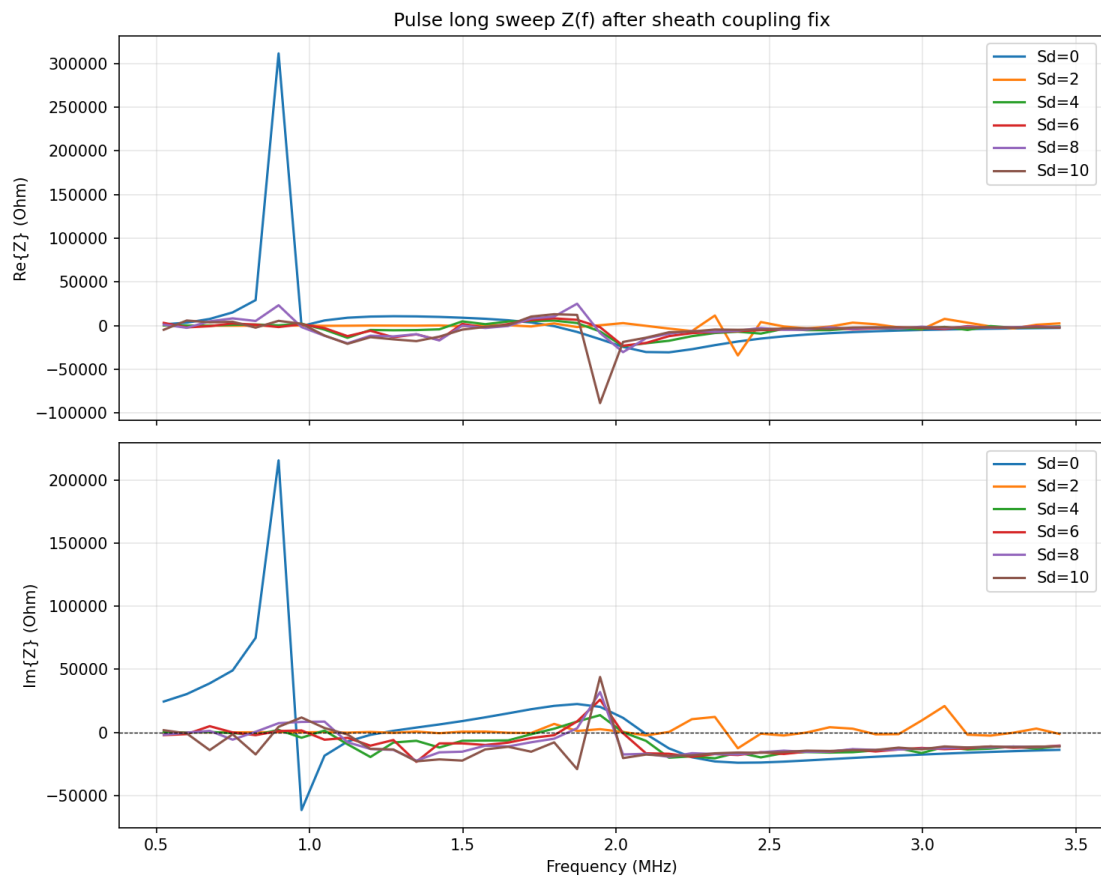


Figure 7.2: Post-fix pulse FFT impedance. Curves separate with S_d ; $S_d = 10$ is capacitive through 1.7–1.9 MHz, matching the CW confirm.

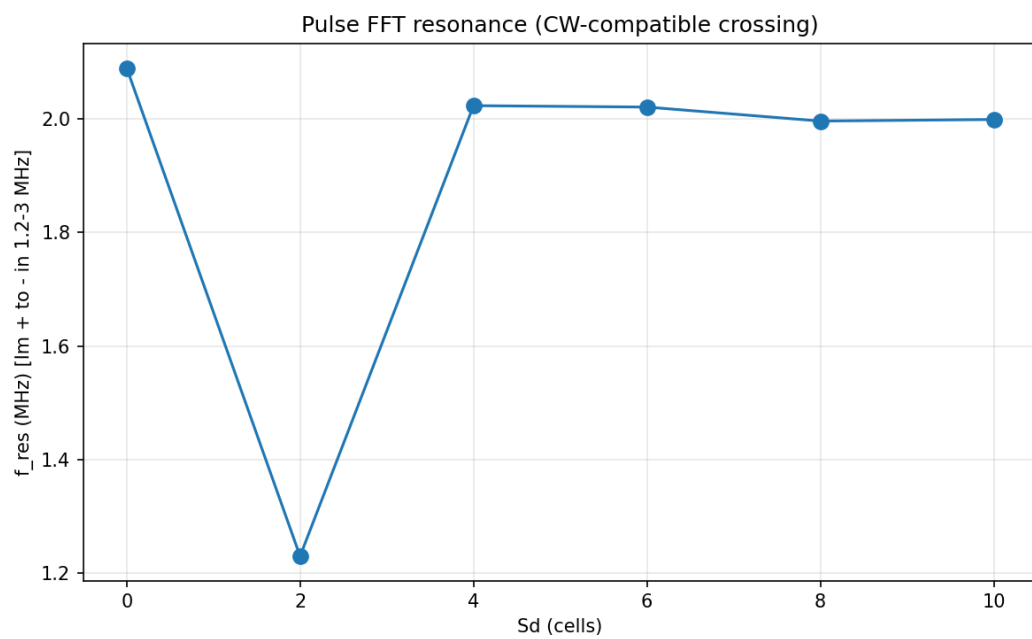


Figure 7.3: Pulse-FFT resonance versus S_d . Frequency resolution is too coarse for a Tu curve; CW remains the preferred f_{res} measurement.

Chapter 8

Dense CW Tu Sweep

8.1 Campaign setup

The dense CW grid was the post-fix measurement intended to produce a clean $f_{\text{res}}(S_d)$ curve. Driver: `scripts/run_sheath_cw_tu.ps1`. Output: `results/sheath_cw_tu/` (voltage/current only; no field dumps).

- $S_d \in \{0, 2, 4, 6, 8, 10\}$.
- Drive frequencies: 1.50, 1.60, 1.70, 1.75, 1.80, 1.85, 1.90, 2.00, 2.10, 2.20, 2.30 MHz (66 cases).
- $f_p = 2$ MHz, template `sheath_sine_vc.str`, `MaxParallel = 3`, `OMP = 4`.
- Started 13 August 2026 12:48; all 66 cases finished by 18 August 2026.

Analyzer: `scripts/analyze_cw_tu_partial.py` (last-50% phasor Z , same convention as the f_p -screen; series resonance is the first $\text{Im}\{Z\} + \text{to} - \text{crossing}$). Tables and figures: `analysis/data/cw_tu*.txt`, `analysis/figures/cw_tu*.png`.

8.2 Campaign status

As of 18 August 2026: **66 complete, 0 running, 0 pending**, 0 failed. Every planned cell in Table 8.1 finished with a full `data.vc` file.

The $S_d = 0$, 1.50 MHz case was skipped once by the orchestrator on restart; its file is smaller than the others and lacks a `DONE` line, but the phasor at 1.50 MHz is consistent with the rising inductive branch below the crossing and is included in the plots.

Table 8.1: CW Tu case status (18 August 2026). All cells complete.

S_d	1.50	1.60	1.70	1.75	1.80	1.85	1.90	2.00	2.10	2.20	2.30
0	C	C	C	C	C	C	C	C	C	C	C
2	C	C	C	C	C	C	C	C	C	C	C
4	C	C	C	C	C	C	C	C	C	C	C
6	C	C	C	C	C	C	C	C	C	C	C
8	C	C	C	C	C	C	C	C	C	C	C
10	C	C	C	C	C	C	C	C	C	C	C

8.3 Impedance versus frequency

Figure 8.1 and Table 8.2 show the full-grid phasor results. Unlike the pre-fix f_p -screen, curves for different S_d are well separated across 1.50–2.30 MHz.

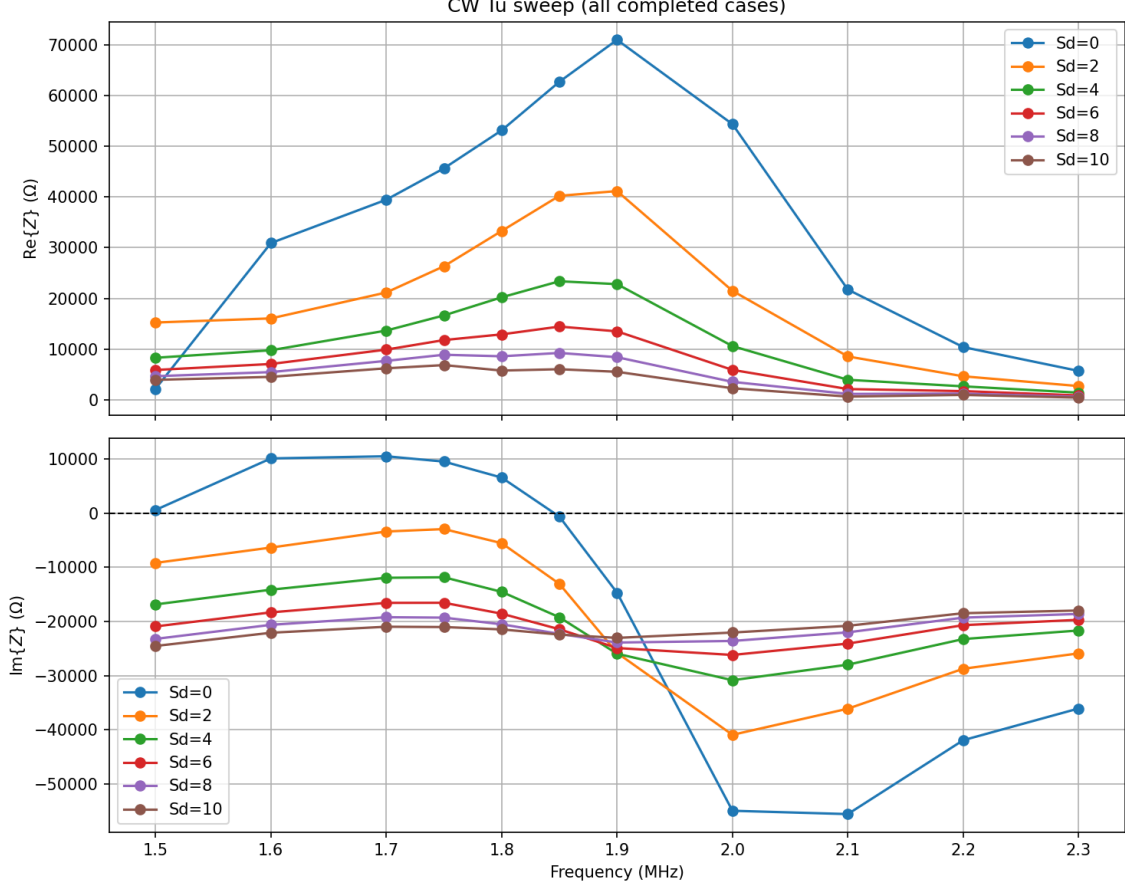


Figure 8.1: CW Tu sweep, all 66 cases. $S_d = 0$ crosses $\text{Im}\{Z\} = 0$ near 1.85 MHz and is capacitive above ~ 2.0 MHz. Every $S_d \geq 2$ curve stays capacitive ($\text{Im}\{Z\} < 0$) across the entire band.

8.4 Series resonance $f_{\text{res}}(S_d)$

Linear interpolation of the $S_d = 0$ + to – crossing between 1.80 and 1.85 MHz gives

$$f_{\text{res}}(S_d = 0) \approx 1.846 \text{ MHz.} \quad (8.1)$$

That matches the post-fix confirm (sign change between 1.7 and 1.9 MHz) and is slightly above the coarse pre-fix 1.7/1.9 interpolation (~ 1.775 MHz).

For $S_d = 2, 4, 6, 8, 10$, $\text{Im}\{Z\} < 0$ at *every* frequency in 1.50–2.30 MHz. No + to – crossing appears anywhere in the grid for $S_d > 0$. Any series crossing for those cases therefore lies *below* 1.50 MHz—the opposite direction from Tu [1], which predicts f_{res} moving *upward* with S_d toward the free-space value. Figure 8.2 contains only the $S_d = 0$ point.

Table 8.2: Selected phasor impedances (Ω). Full table: `analysis/data/cw_tu_summary.txt`.

f (MHz)	S_d	$\text{Re}\{Z\}$	$\text{Im}\{Z\}$	note
1.50	0	2130	+585	inductive shoulder
1.85	0	62775	-640	just past crossing
2.00	0	54406	-54920	capacitive above f_{res}
1.50	2	15227	-9181	capacitive at band edge
2.00	2	21484	-40907	still capacitive
1.70	10	6182	-20959	matches confirm-Z
2.00	10	2238	-22032	$ Z $ lowest in band

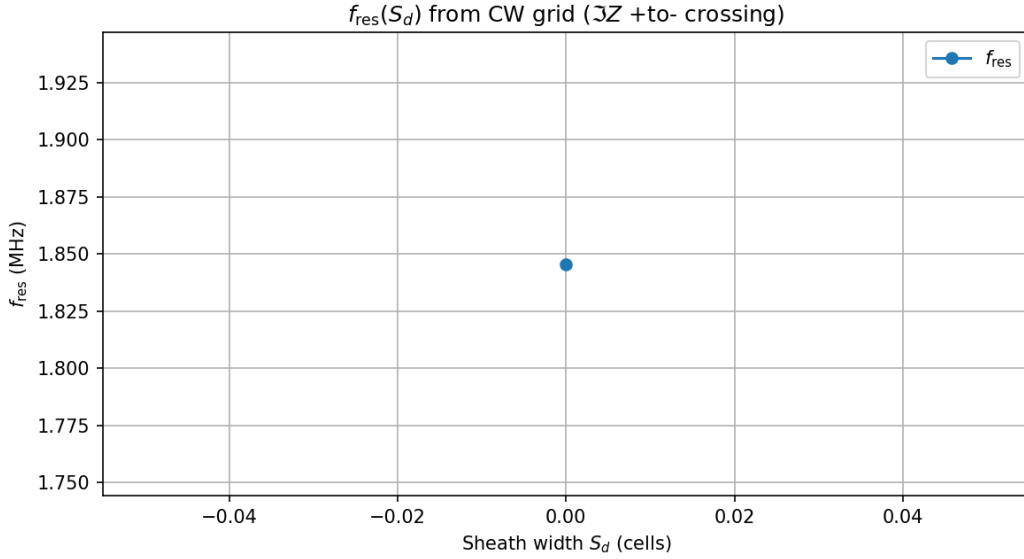


Figure 8.2: $f_{\text{res}}(S_d)$ from the full CW grid ($\text{Im}\{Z\}$ + to - crossing). Only $S_d = 0$ has a crossing in 1.50–2.30 MHz.

Table 8.3 summarizes the resonance extraction. The high-frequency extension (1.90–2.30 MHz) rules out the provisional hypothesis from the partial run that a Tu-like upward crossing might appear above 1.9 MHz for large S_d : $S_d = 10$ remains strongly capacitive at 2.0 and 2.3 MHz.

8.5 What succeeded

- Coupling is visible in CW at every $S_d > 0$: $Z(f)$ is no longer independent of sheath width.
- The $S_d = 0$ plasma-loaded resonance is resolved to ~ 50 kHz spacing near f_p .
- The full grid confirms the post-fix confirm-Z picture: $S_d = 0$ is inductive below ~ 1.85 MHz; $S_d = 10$ is capacitive everywhere in the band.

8.6 What did not match Tu (2008)

The Tu observable is the *upward shift* of f_{res} with S_d . That shift was not reproduced:

Table 8.3: Series resonance from $\text{Im}\{Z\}$ + to – crossings.

S_d	f_{res}	note
0	1.846 MHz	crossing in 1.50–2.30 MHz band
2	—	all $\text{Im}\{Z\} < 0$; crossing at or below 1.50 MHz
4	—	all $\text{Im}\{Z\} < 0$; crossing at or below 1.50 MHz
6	—	all $\text{Im}\{Z\} < 0$; crossing at or below 1.50 MHz
8	—	all $\text{Im}\{Z\} < 0$; crossing at or below 1.50 MHz
10	—	all $\text{Im}\{Z\} < 0$; crossing at or below 1.50 MHz

- Only $S_d = 0$ yields a measurable in-band series crossing.
- Wider sheaths add capacitive loading ($|Z|$ falls with S_d at fixed f) but do not move a + to – crossing upward through 1.50–2.30 MHz.
- If the same crossing definition applies to $S_d \geq 2$, the feature has moved *downward* relative to $S_d = 0$, or the feed impedance is dominated by sheath capacitance without a clear in-band LC crossing.

Possible follow-ups (Chapter 9): extend the CW grid below 1.50 MHz for $S_d \geq 2$; compare against free-space f_{res} on the same grid; revisit whether the + to – crossing is the right marker when a wide vacuum sheath makes the feed uniformly capacitive in the band.

Chapter 9

Status and Next Steps

9.1 Where the campaign stands

The work from February through 18 August 2026 is one antenna-in-plasma story on a single dipole grid.

1. **February baseline.** Free-space versus plasma impedance is measurable. Same geometry as all later sheath runs.
2. **May pulse S_d sweep.** Failed as a measurement (no usable f_{res} ; GHz FFT artifacts). Independently, the sheath was not on the wire.
3. **Low-frequency CW.** Measurement method works; frequencies sit below resonance; Z still independent of S_d .
4. **CW f_p -screen.** Measurement success ($f_{\text{res}} \approx 1.775$ MHz) and physics failure ($Z(S_d = 0) \equiv Z(S_d = 10)$ to $\sim 0.003\%$). Negative control, keep the folder.
5. **Coupling fix.** Sheath had been seeded from the plasma-on mask before PEC existed; depletion sat at the domain edge. PEC-seeded `ApplySheath` after `setup2` puts the hole next to the wire.
6. **Post-fix confirm.** $|\Delta Z|/|Z| \sim 90\text{--}110\%$ at 1.7/1.9 MHz. Pulse FFT now sees S_d but is too coarse for a Tu curve.
7. **Dense CW (complete).** All 66 cases finished. $f_{\text{res}}(S_d = 0) \approx 1.846$ MHz. $S_d \geq 2$ is capacitive at every point in 1.50–2.30 MHz. The Tu upward shift of f_{res} with S_d was not reproduced with the + to $-\text{Im}\{Z\}$ crossing definition.

9.2 Open items

1. Extend the CW grid *below* 1.50 MHz for $S_d \geq 2$ to locate any + to $-$ crossings outside the August band, or adopt an alternative resonance marker when the feed is uniformly capacitive.
2. Compare $f_{\text{res}}(S_d = 0)$ and any low-frequency crossings to the free-space resonance on the same dipole grid.
3. Optional: longer muted pulse ($\text{MaxIter} \gtrsim 10^6$) if additional frequency resolution is needed outside CW.

4. Optional: re-run the f_p -screen with the fixed physics as a published before/after pair. Confirm-Z and the CW Tu grid already show coupling; this is documentation, not debugging.
5. Software not in this report's critical path: GPU, MPI, broader unit-test coverage.

9.3 Data map

Table 9.1: Primary data and writeups for each phase.

Phase	Results	Notes / figures
Feb baseline	<code>results/freespace_test</code> , <code>ionoplasma_*</code>	Ch. 2
Pulse S_d	<code>results/sheath_sweep</code> , <code>sheath_long</code>	<code>analysis/sheath_coupling_findings.m</code>
Narrow / 2D CW	<code>results/sheath_narrow_band</code> , <code>sheath_2d_sweep</code>	Ch. 4
f_p -screen	<code>results/sheath_fp_screen</code>	pre-fix negative control
N_0 dump	<code>results/n0_diag</code>	<code>analysis/figures/n0_line_*.png</code>
Confirm	<code>results/sheath_confirm_z</code> , <code>sheath_pulse_long</code>	Ch. 7
CW Tu	<code>results/sheath_cw_tu</code>	Ch. 8; <code>analysis/data/cw_tu_*.txt</code>

The coupling-fix narrative in `analysis/sheath_coupling_findings.md` remains the detailed laboratory note; this report is the chronological account intended for sharing as PDF.

Bibliography

- [1] W. Tu, J.-L. Pathak, and R. F. Benson, *Plasma sheath structures around a radio frequency antenna*, J. Geophys. Res., 2008. A local copy is in docs/JGR2008 - Tu - Plasma sheath structures around a radio frequency antenna.pdf.
- [2] J. D. Ward, *Plasma Frequency Probe Design and Implementation*, Master's thesis, Utah State University, 2006.